

Continental Express Flight 2574

Flight model: Embraer 120 Brasilia
Date: September 11, 1991

From: Laredo, US
To: Houston, US

Problem

The plane had reached the cruising altitude. Then, suddenly the plane falls from the sky. People on the ground had noticed that there was fire in the right wing of the plane and the wing was not intact. Finally, crashes.

Investigation

In Embraer 120, the **horizontal stabilizer** sit on top of the **vertical stabilizer**. This is called a **T-tail**. The horizontal stabilizer were found 200 meters from the crash site which clearly indicate an in flight breakup. It was also found the left leading edge of the horizontal stabilizer is also missing as a whole which is usual as it is possible to find one piece missing and not as whole. This leading edge is important for a smooth flow of air around the stabilizer. When it was found, the bottom of the leading edge had stress marks clearly showing that it was subjected to lots of stress but the top of the leading edge looked clean with no marks or sign of stress which indicate that the top of the leading edge had no screws or fasteners. It must have been the first piece to fall of the plane. Simulation of losing the leading edge show that it may lead the plane in a nose dive. The Continental express airline was preparing its fleet for the winter by checking the de-icing equipment, which includes the leading edge which has a boot inside it to break ice accumulated on the horizontal stabilizers. It was found that the maintenance inspector had offered to help the maintenance mechanics by removing the top screws of the leading edge stabilizer. But, the mechanics of the next shift had only worked on the right of the horizontal stabilizer and not the left leaving the screws unscrewed. The inspector had to monitor the working of the mechanics and should not have done the role of the mechanics which directs to the principal that when one does the work the best of the best (here, the inspector) should monitor the work and should not have worked. There was no paper work done when they moved from one shift to another which was mandatory to make sure things were done properly. The inspector who had written a shift report was not clear on what work that was exactly performed.

Changes that were brought after the incident

The FAA blamed the airline for the staff to not to perform proper paper work after the maintenance shift and not following proper procedure. After the crash airline put much more emphasis on the safety procedure to be followed after the maintenance work is completed. They also use computer systems to track the work of mechanics. It also increases emphasis on the employee interaction among their college to convey any update on the work regardless on how small the work is. Small deviation by many people had lead to the cause of the major problem in the plane.

Reference

1. The Fatal Mistake on Flight 2574 — Mayday: Air Disaster
<https://www.youtube.com/watch?v=NYtMYeGiTjI>